

from Beginner to Master

DevOps를 위한 **Docker** 가상화

Section 8. Logging and Monitoring

```
book:
  def setData(self, title, price, author):
    self.title = title
    self.price = price
    self.author = author

var fs = require('fs');
fs.readFile('./ONE.txt', '* 1 *',
  function (err, data) {
    console.log(data); // 3
  });
ApplicationDelegate > {
```

```
public static void main(String[] args)

<servlet>
  <servlet-name>BoardController</servlet-name>
  <servlet-class>com.joneconsulting.controller.BoardCont
  <init-param>
    <param-name>user_name</param-name>
    <param-value>henmeth Lee</param-value>
  </init-param>
</servlet>
```

Winterface

Dowon Lee

수강생 19K 강의평점 4.8(1K)



멘토링 ON

멘토링 설정

- 홈
- 강의
- 로드맵
- 수강평
- 게시글
- 블로그

수정하기

강의 전체 6



[개정판] 웹 애플리케이션 개발을 위한 IntelliJ IDEA 설정

▶ 학습중

★ 0강 / 25강 (0%)

818명



멀티OS 사용을 위한 가상화 환경 구축 가이드 (Docker + Kubernetes)

▶ 학습중

★ 0강 / 16강 (0%)

924명



Jenkins를 이용한 CI/CD Pipeline 구축

▶ 학습중

★ 81강 / 83강 (98%)

독점 2709명

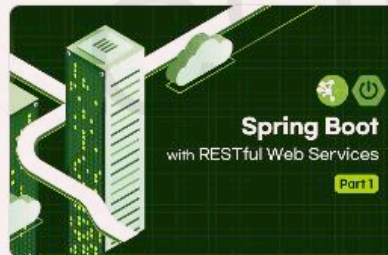


Spring Cloud로 개발하는 마이크로서비스 애플리케이션(MSA)

▶ 학습중

★ 156강 / 156강 (100%)

독점 5531명



Spring Boot를 이용한 RESTful Web Services 개발

▶ 학습중

★ 3강 / 48강 (6%)

독점 3862명



[구버전] 웹 애플리케이션 개발을 위한 IntelliJ IDEA 설정 (2020 ver.)

▶ 학습중

★ 14강 / 14강 (100%)

4813명

이 되었던 때가 있었습니
"IT 엔지니어"라고 적고

을 가지고 있습니다. 누구
사람입니다. 개발자로서, 강
지만, 그래도, 남들보다 조

ecture, Blockchain,

- Section 0: Introduction to Cloud Native Technologies
- Section 1: Docker Essentials - Container
- Section 2: Docker Essentials - Image
- Section 3: Docker Network and Storage
- Section 4: Building and Managing Containerized Application
- Section 5: Container Orchestration
- Section 6: Continuous Integration and Continuous Deployment
- Section 7: Docker Security
- **Section 8: Logging and Monitoring**
- Section 9: Advanced Docker Usage
- Section 10: Capstone Project

Section 8.

Logging and Monitoring

- Docker Container Logging
- Docker Container Monitoring



Docker Container Logging

Logging

- 도커 내부에서 발생하는 로그 메시지 확인

| 도커 실행 시 또는 실행 이후에 발생 되는 로그 메시지 출력

```
$ docker logs <Container ID or Name>
```

| mariadb 실행에 대한 로그 확인

```
$ docker run -d -p 13306:3306 \  
    -e MARIADB_ALLOW_EMPTY_ROOT_PASSWORD=true \  
    --name mariadb1 \  
    mariadb:latest
```

```
$ docker logs mariadb1
```

Docker Container Logging

Logging (Linux)

- 컨테이너 로그 파일 확인

- | docker inspect → "LogPath": "/var/lib/docker/containers/~
 - | /var/lib/docker/containers/\${CONTAINER_ID}/\${CONTAINER_ID}-json.log

```
[ec2-user@ip-172-31-5-111 ~]$ docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
7aa484a25939	mariadb:latest	"docker-entrypoint.s..."	7 minutes ago	Up 7 minutes	0.0.0.0:13306->3306

```
[ec2-user@ip-172-31-5-111 ~]$ sudo ls -al /var/lib/docker/containers/7aa484a2593964038ce8c5e7bb86494bf23f4040293e3ab9ae17de6680c98bab-
total 40
drwx--x--- 4 root root  237 Apr 21 14:29 .
drwx--x--- 3 root root   78 Apr 21 14:29 ..
-rw-r----- 1 root root 14313 Apr 21 14:29 7aa484a2593964038ce8c5e7bb86494bf23f4040293e3ab9ae17de6680c98bab-json.log
```

```
$ sudo cat /var/lib/docker/containers/7aa484a~/7aa484a~-json.log
```

Docker Container Logging

Logging (Linux)

- 컨테이너 로그 파일 → syslog

| syslog → 유닉스 계열 운영체제에서 로그 수집하는 도구 (커널, 보안, 애플리케이션 로그 등)

```
$ docker run -d -p 13306:3306 --log-driver syslog  
-e MARIADB_ALLOW_EMPTY_ROOT_PASSWORD=true \  
--name mariadb1 \  
mariadb:latest
```

```
[ec2-user@ip-172-31-5-111 ~]$ docker run -d --rm --log-driver=syslog -p 13306:3306 -e MARIADB_ALLOW_EMPTY_ROOT_PASSWORD=true --name mariadb1 mariadb:latest  
c68c243acbb35b9c853e4947b556247857469f36a56c0cd02d8a27bb5366e79c  
[ec2-user@ip-172-31-5-111 ~]$ journalctl -u docker.service | grep c68c  
Apr 21 14:48:09 ip-172-31-5-111.ap-northeast-2.compute.internal dockerd[3101]: time="2024-04-21T14:48:09.291290636Z" level=info msg="Configured log driver does not support reads, enabling local file cache for container logs" container=c68c243acbb35b9c853e4947b556247857469f36a56c0cd02d8a27bb5366e79c driver=syslog  
Apr 21 14:48:09 ip-172-31-5-111.ap-northeast-2.compute.internal c68c243acbb3[3101]: 2024-04-21 14:48:09+00:00 [Note] [Entrypoint]: Entrypoint script for MariaDB Server 1:11.3.2+maria~ubu2204 started.  
Apr 21 14:48:09 ip-172-31-5-111.ap-northeast-2.compute.internal c68c243acbb3[3101]: 2024-04-21 14:48:09+00:00 [Warn] [Entrypoint]: /sys/fs/cgroup/freezer:/docker/c68c243acbb35b9c853e4947b556247857469f36a56c0cd02d8a27bb5366e79c  
Apr 21 14:48:09 ip-172-31-5-111.ap-northeast-2.compute.internal c68c243acbb3[3101]: 10:devices:/docker/c68c243acbb35b9c853e4947b556247857469f36a56c0cd02d8a27bb5366e79c
```


Docker Container Logging

Logging (Windows)

- 컨테이너 로그 파일 확인

- | docker inspect → "LogPath": "/var/lib/docker/containers/~
| /var/lib/docker/containers/\${CONTAINER_ID}/\${CONTAINER_ID}-json.log

- Windows에서는 /var/lib/docker/containers/~ 의 내용 확인 어려움

- | /var/lib/docker/container를 Volume mount로 연결한 Linux Container에서 내용 확인 가능

```
C:\> docker run -v /:/data -it ubuntu /bin/bash
```

```
# chroot /data
```

```
# cd /var/lib/docker/containers
```

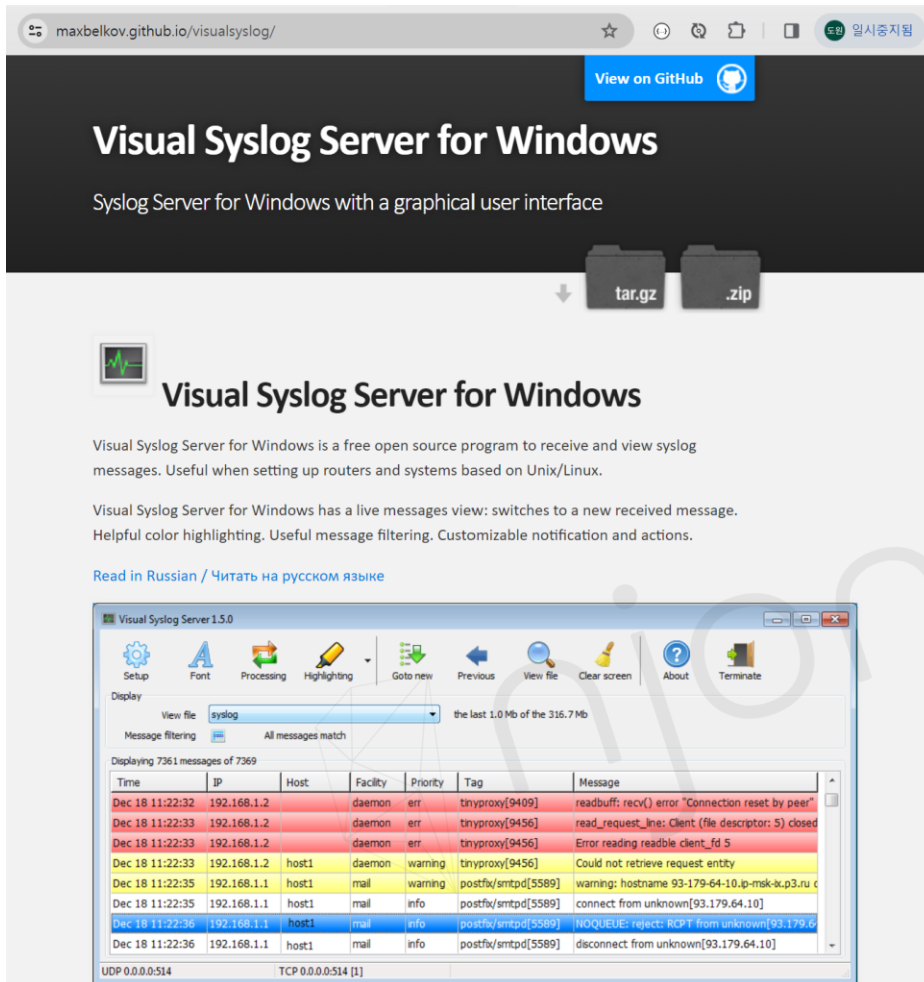
```
# ls -al
```

```
drwx--x---  4 root root  4096 Apr 22 14:13 9d19a5067fcbdb80d634b8a5e343d150ec9a7b09aa15d4da9fa291c7f40a769e
drwx--x---  4 root root  4096 Apr 22 14:13 b4526e85340fb2d2fad1cda441b7f332baa5e5bb0a3b1fb13b4ebb0b746ee3ac
drwx--x---  4 root root  4096 Apr 22 14:13 b89558bb32a825b1b317f28b93e51f5c504a3a96edc2d7a2b0202c3c49635826
drwx--x---  4 root root  4096 Apr 22 14:13 b99e9ba87df48130e64d5be668f3747c275534d867fbd61be6c226aa9a855316
drwx--x---  4 root root  4096 Apr 22 14:13 bfef3c20b11c50c4ac418e96c3bf89f0519c2041c50953e8e00172f5f723e218
```


Syslog Server for Windows

Visual Syslog

- <https://maxbelkov.github.io/visualsyslog/>



The screenshot shows the GitHub page for Visual Syslog Server for Windows. The page has a dark header with the title "Visual Syslog Server for Windows" and a subtitle "Syslog Server for Windows with a graphical user interface". Below the header, there are download buttons for "tar.gz" and ".zip". The main content area features a green heart icon and the title "Visual Syslog Server for Windows". Below this, there is a paragraph describing the program: "Visual Syslog Server for Windows is a free open source program to receive and view syslog messages. Useful when setting up routers and systems based on Unix/Linux." Another paragraph mentions features: "Visual Syslog Server for Windows has a live messages view: switches to a new received message. Helpful color highlighting. Useful message filtering. Customizable notification and actions." There is a link to "Read in Russian / Читать на русском языке". At the bottom, there is a screenshot of the Visual Syslog Server 1.5.0 application window.

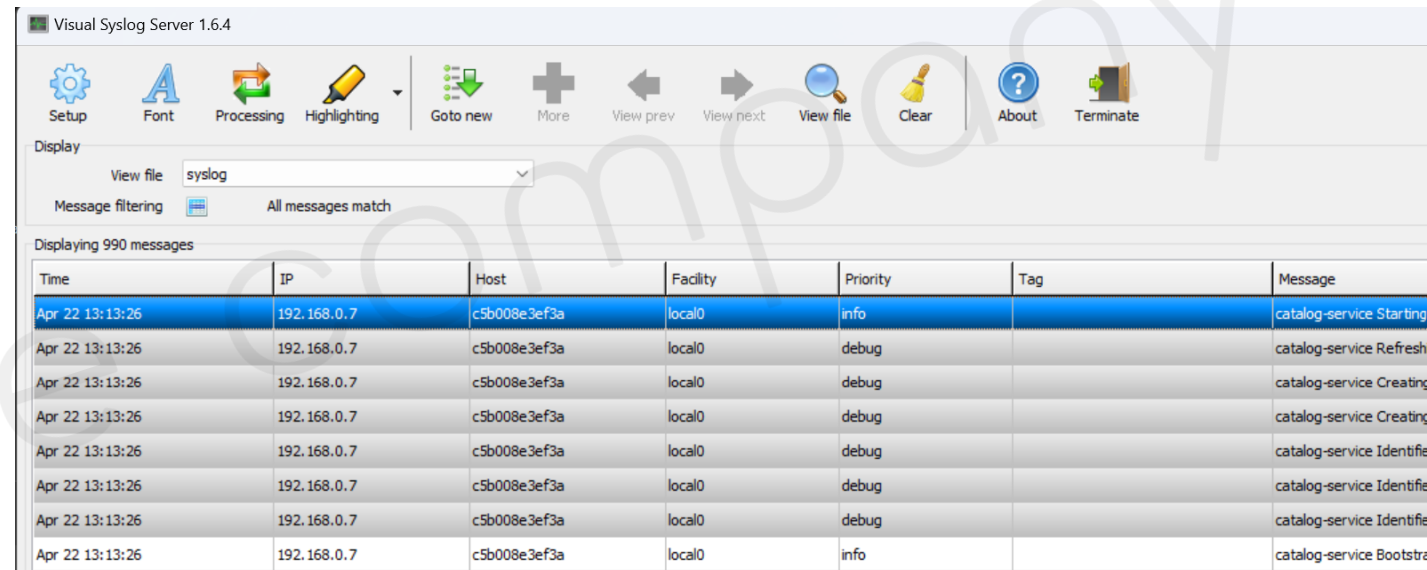
Visual Syslog Server 1.5.0

Display: View file: syslog, Message filtering: All messages match, the last 1.0 Mb of the 316.7 Mb

Displaying 7361 messages of 7369

Time	IP	Host	Facility	Priority	Tag	Message
Dec 18 11:22:32	192.168.1.2	daemon	err	tinyproxy[9409]	readbuff: recv() error "Connection reset by peer"	
Dec 18 11:22:33	192.168.1.2	daemon	err	tinyproxy[9456]	read_request_line: Client (file descriptor: 5) closed	
Dec 18 11:22:33	192.168.1.2	daemon	err	tinyproxy[9456]	Error reading readable client_fd 5	
Dec 18 11:22:33	192.168.1.2	host1	daemon	warning	tinyproxy[9456]	Could not retrieve request entity
Dec 18 11:22:35	192.168.1.1	host1	mail	warning	postfix/smtpd[5589]	warning: hostname 93-179-64-10.jp-msk-ix.p3.ru c
Dec 18 11:22:35	192.168.1.1	host1	mail	info	postfix/smtpd[5589]	connect from unknown[93.179.64.10]
Dec 18 11:22:36	192.168.1.1	host1	mail	info	postfix/smtpd[5589]	NOQUEUE: reject: RCPT from unknown[93.179.6
Dec 18 11:22:36	192.168.1.1	host1	mail	info	postfix/smtpd[5589]	disconnect from unknown[93.179.64.10]

UDP 0.0.0.0:514 TCP 0.0.0.0:514 [1]



The screenshot shows the Visual Syslog Server 1.6.4 application window. The interface includes a menu bar with options: Setup, Font, Processing, Highlighting, Goto new, More, View prev, View next, View file, Clear, About, and Terminate. Below the menu bar, there is a "Display" section with a "View file" dropdown set to "syslog" and a "Message filtering" section set to "All messages match". The main area displays "Displaying 990 messages" and a table of syslog messages.

Time	IP	Host	Facility	Priority	Tag	Message
Apr 22 13:13:26	192.168.0.7	c5b008e3ef3a	local0	info		catalog-service Starting
Apr 22 13:13:26	192.168.0.7	c5b008e3ef3a	local0	debug		catalog-service Refresh
Apr 22 13:13:26	192.168.0.7	c5b008e3ef3a	local0	debug		catalog-service Creating
Apr 22 13:13:26	192.168.0.7	c5b008e3ef3a	local0	debug		catalog-service Creating
Apr 22 13:13:26	192.168.0.7	c5b008e3ef3a	local0	debug		catalog-service Identify
Apr 22 13:13:26	192.168.0.7	c5b008e3ef3a	local0	debug		catalog-service Identify
Apr 22 13:13:26	192.168.0.7	c5b008e3ef3a	local0	debug		catalog-service Identify
Apr 22 13:13:26	192.168.0.7	c5b008e3ef3a	local0	info		catalog-service Bootstr

Docker Container Logging

Logging (Windows)

■ Syslog 서버 설치 필요

- Windows에서는 Syslog를 사용할 수 없기 때문에, 별도의 Syslog Server 설치

```
$ docker run -d -p 13306:3306 --log-driver syslog \
    --log-opt syslog-address=udp://192.168.0.7:514 \
    -e MARIADB_ALLOW_EMPTY_ROOT_PASSWORD=true \
    --name mariadb1 \
    mariadb:latest
```

| syslog 파일 위치 → C:\W<USER_HOME>\WAppData\WLocal\Wvisualsyslog\Wsyslog

```
syslog
C: > Users > zitan > AppData > Local > visualsyslog > syslog
1 192.168.0.7 Apr 22 11:54:13 daemon info c7507dee37c6[1298] 2024-04-22 11:54:13+00:00 [Note] [Entrypoint]: Entrypoint script for MariaDB Server 1:10.11.2+maria~ubu2204 started.
2 192.168.0.7 Apr 22 11:54:13 daemon info c7507dee37c6[1298] 2024-04-22 11:54:13+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
3 192.168.0.7 Apr 22 11:54:13 daemon info c7507dee37c6[1298] 2024-04-22 11:54:13+00:00 [Note] [Entrypoint]: Entrypoint script for MariaDB Server 1:10.11.2+maria~ubu2204 started.
4 192.168.0.7 Apr 22 11:54:13 daemon info c7507dee37c6[1298] 2024-04-22 11:54:13+00:00 [Note] [Entrypoint]: Initializing database files
5 192.168.0.7 Apr 22 11:54:13 daemon err c7507dee37c6[1298] 2024-04-22 11:54:13 0 [Warning] mariabdd: io_uring_queue_init() failed with ENOSYS: check seccomp filters, and the k
6 192.168.0.7 Apr 22 11:54:13 daemon err c7507dee37c6[1298] 2024-04-22 11:54:13 0 [Warning] InnoDB: liburing disabled: falling back to innodb_use_native_aio=OFF
7 192.168.0.7 Apr 22 11:54:16 daemon info c7507dee37c6[1298] PLEASE REMEMBER TO SET A PASSWORD FOR THE MariaDB root USER !
8 192.168.0.7 Apr 22 11:54:16 daemon info c7507dee37c6[1298] To do so, start the server, then issue the following command:
9 192.168.0.7 Apr 22 11:54:16 daemon info c7507dee37c6[1298] '/usr/bin/mariadb-secure-installation'
10 192.168.0.7 Apr 22 11:54:16 daemon info c7507dee37c6[1298] which will also give you the option of removing the test
11 192.168.0.7 Apr 22 11:54:16 daemon info c7507dee37c6[1298] databases and anonymous user created by default. This is
12 192.168.0.7 Apr 22 11:54:16 daemon info c7507dee37c6[1298] strongly recommended for production servers.
13 192.168.0.7 Apr 22 11:54:16 daemon info c7507dee37c6[1298] See the MariaDB Knowledgebase at https://mariadb.com/kb
```

Docker Container Logging

Logging (Windows)

■ Syslog 서버 설치 필요

- my-backend 예제 사용 → section8-win branch

■ <https://github.com/joneconsulting/devops-docker/tree/main/my-backend>

```
$ docker-compose --f docker-compose-win.yml up -d
```

```
version: '3.7'
services:
  my-backend:
    image: edowon0623/catalog-service:syslog-ng-v1.3
    environment:
      - spring.datasource.url=jdbc:mariadb://my-db:3306/mydb
      - spring.datasource.password=
      - DOCKER-COMPOSE-SYSLOG-ADDRESS=192.168.0.7
      - DOCKER-COMPOSE-SYSLOG-PORT=514
    ports:
      - 8088:8088
    depends_on:
      - my-db
    networks:
      - my-network
```

■ Windows PC ip-address

```
<configuration debug="true" scan="true" scanPeriod="30 seconds">
  <property name="SYSLOG-ADDRESS" value="${DOCKER-COMPOSE-SYSLOG-ADDRESS}"/>
  <property name="SYSLOG-PORT" value="${DOCKER-COMPOSE-SYSLOG-PORT}"/>

  <appender name="SYSLOG" class="ch.qos.logback.classic.net.SyslogAppender">
    <syslogHost>${SYSLOG-ADDRESS}</syslogHost>
    <facility>LOCAL0</facility>
    <port>${SYSLOG-PORT}</port>
    <throwableExcluded>true</throwableExcluded>
    <suffixPattern>catalog-service %m thread:%t priority:%p category:%c
    exception:%exception</suffixPattern>
  </appender>
```

Docker Container Logging

Logging (Windows)

- Spring Boot Application + MariaDB → Syslog

Visual Syslog Server 1.6.4

Setup Font Processing Highlighting Goto new More View prev View next View file Clear About Terminate

Display

View file: syslog [the last 1.0 Mb of the 1.0 Mb]

Message filtering: ☒ Text contains "called"

Displaying 3 messages of 3049

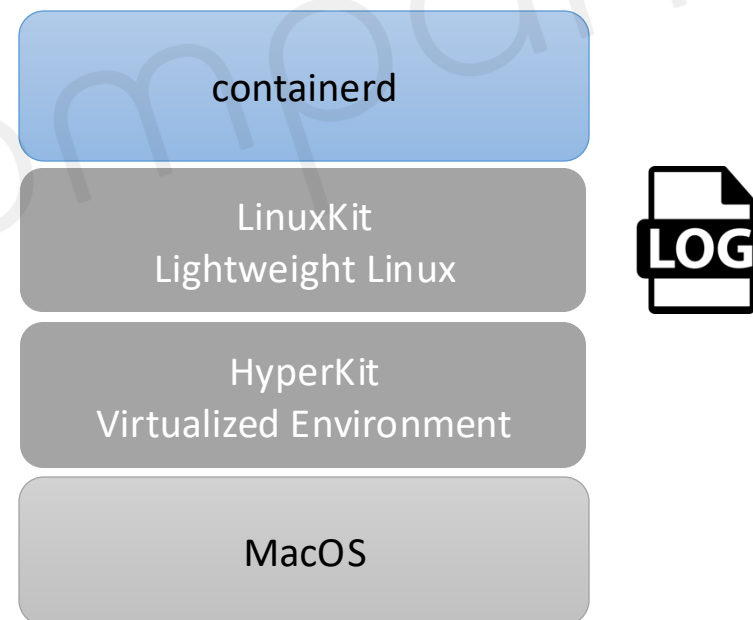
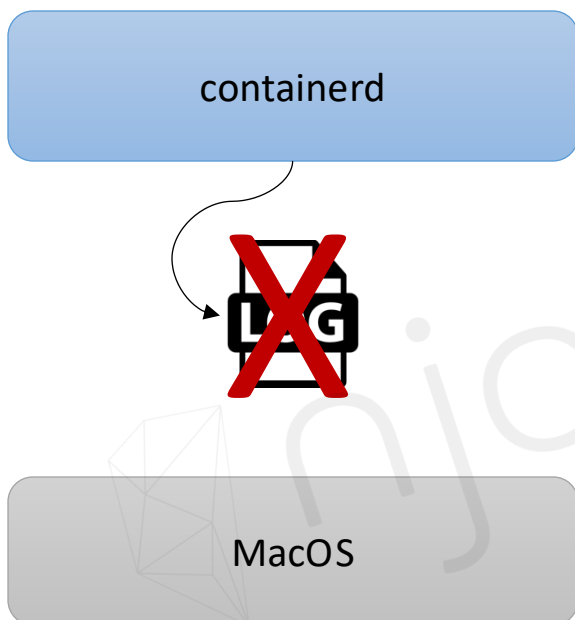
Time	IP	Host	Facility	Priority	Tag	Message
Apr 22 21:48:08	192.168.0.7	rog-flow	local0	debug		catalog-service called catalogs thread:http-nio-8088-exec-2 priority:DEBUG category:com.example.catalogservice.controller.CatalogController exception:
Apr 22 21:52:59	192.168.0.7	rog-flow	local0	debug		catalog-service called catalogs thread:http-nio-8088-exec-5 priority:DEBUG category:com.example.catalogservice.controller.CatalogController exception:
Apr 22 13:07:19	192.168.0.7	ac387661012e	local0	debug		catalog-service [called catalogs] thread:http-nio-8088-exec-2 priority:DEBUG category:com.example.catalogservice.controller.CatalogController exception:

Docker Container Logging

Logging (MacOS)

- 컨테이너 로그 파일 확인

- | `docker inspect` → "LogPath": `"/var/lib/docker/containers/~`
- | `/var/lib/docker/containers/${CONTAINER_ID}/${CONTAINER_ID}-json.log`



Docker Container Logging

Logging (MacOS)

- MacOS Apple chip 에서는 /var/lib/docker/containers/~ 의 내용 확인 어려움

<https://gist.github.com/BretFisher/5e1a0c7bcc4c735e716abf62afad389>

```
$ docker run -it --rm --privileged --pid=host justincormack/nsenter1  
  
# cd /var/lib/docker/containers  
  
# ls -al
```

```
▶ docker run -it --rm --privileged --pid=host justincormack/nsenter1  
~ # cd /var/lib/docker/containers  
/var/lib/docker/containers # ls -al  
total 16  
drwx--x---  4 root    root    4096 Apr 25 08:55 .  
drwx--x--- 12 root    root    4096 Apr 25 07:51 ..  
drwx--x---  4 root    root    4096 Apr 25 08:41 b8423bbc4892bcaa0dbce80562f8038f4ab0ca4f9f0d23206e875eb0e4368cdf  
drwx--x---  4 root    root    4096 Apr 25 08:55 c9fbbc4457fde5b6c3df3b71a13fdf33e7e599c32abaec519d34d503e89a304d
```

Docker Container Logging

Logging (MacOS)

- 컨테이너 로그 파일 → syslog

| syslog → 유닉스 계열 운영체제에서 로그 수집하는 도구 (커널, 보안, 애플리케이션 로그 등)

| <https://github.com/joneconsulting/devops-docker.git> → section8 branch

```
$ cd my-brackend
```

```
$ docker-compose -f docker-compose.yml up syslog-ng -d
```

```
$ docker-compose ps
```

```
services:
  syslog-ng:
    image: linuxserver/syslog-ng:latest
    container_name: syslog-ng
    environment:
      - PUID=1000
      - PGID=1000
      - TZ=Etc/UTC
    volumes:
      - ./syslog-ng/config:/config
      - ./syslog-ng/log:/var/log #optional
    ports:
      - 514:5514/udp
```


Docker Container Logging

Logging (MacOS)

- Mariadb → Syslog

| mariadb container에서 syslog-ng container로 로그 출력

```
$ docker run -d -p 13306:3306 --log-driver syslog \
--log-opt syslog-address=udp://172.17.0.1:514 \
-e MARIADB_ALLOW_EMPTY_ROOT_PASSWORD=true \
--name mariadb1 \
mariadb:latest
```

```
Apr 29 13:30:19 172.20.0.1 bd7663fe5b59[311]: 2024-04-29 13:30:19 0 [Note] InnoDB: Buffer pool(s) load completed at 240429 13:30:19
Apr 29 13:30:19 172.20.0.1 bd7663fe5b59[311]: 2024-04-29 13:30:19 0 [Note] Server socket created on IP: '0.0.0.0'.
Apr 29 13:30:19 172.20.0.1 bd7663fe5b59[311]: 2024-04-29 13:30:19 0 [Note] Server socket created on IP: '::'.
Apr 29 13:30:19 172.20.0.1 bd7663fe5b59[311]: 2024-04-29 13:30:19 0 [Note] mariadbd: Event Scheduler: Loaded 0 events
Apr 29 13:30:19 172.20.0.1 bd7663fe5b59[311]: 2024-04-29 13:30:19 0 [Note] mariadbd: ready for connections.
Apr 29 13:30:19 172.20.0.1 bd7663fe5b59[311]: Version: '11.3.2-MariaDB-1:11.3.2+maria~ubu2204' socket: '/run/mysqld/mysqld.sock' port: 3306 mariadb.org binary distribution
Apr 29 22:30:37 192.168.65.1 AddressBookSourceSync[82513]:
Apr 29 22:30:40 192.168.65.1 AddressBookManager[82546]:
Apr 29 22:30:51 192.168.65.1 AddressBookManager[82612]:
Apr 29 22:31:48 192.168.65.1 powerd[99]: [System: PrevIdle PrevDisp DeclUser kDisp]
```

Docker Container Logging

Logging Demo

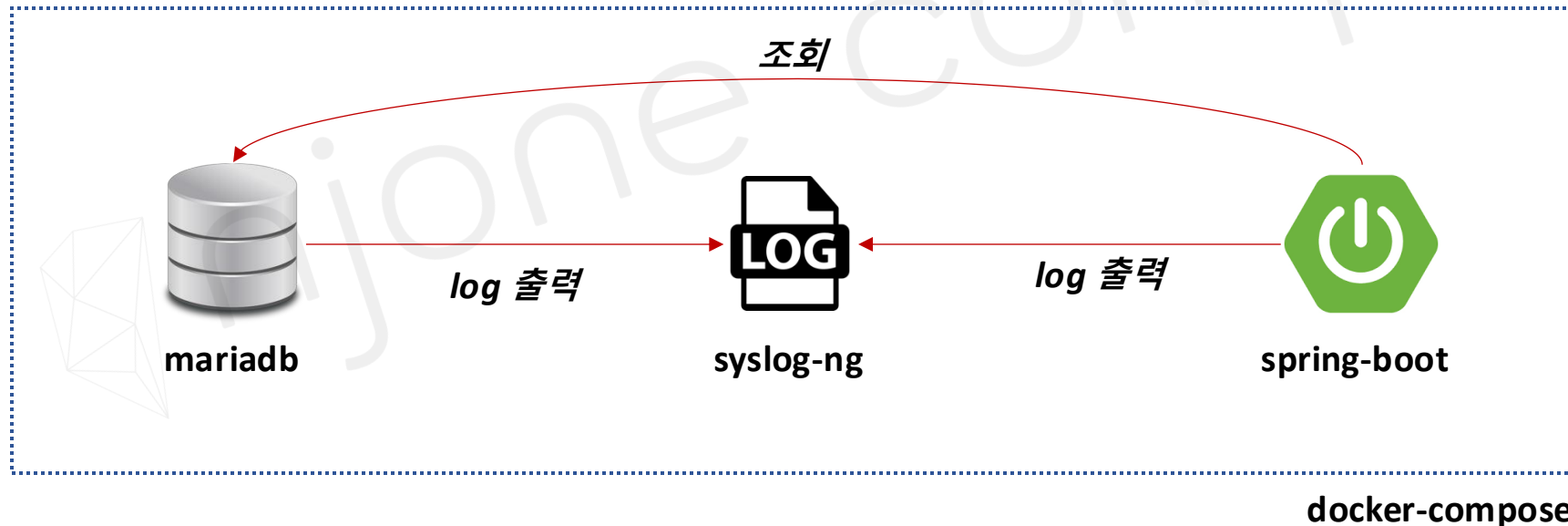
- syslog-ng + mariadb + my-backend

- | docker-compose로 syslog-ng / mariadb / my-backend 서비스 실행

- | <https://github.com/joneconsulting/devops-docker.git> → section8 branch

```
$ cd my-backend
```

```
$ docker-compose -f docker-compose-app.yml up -d
```



Monitoring

- 일반 Monitoring → 서버의 CPU, 메모리 사용량, 디스크 공간 등의 기본 지표를 추적
- 컨테이너 Monitoring → 컨테이너의 생명 주기가 짧고, 여러 개의 컨테이너 관리 필요
- Prometheus
 - | 컨테이너 환경의 애플리케이션을 모니터링 하기 위한 오픈소스 시스템
 - | 여러 컨테이너 상태를 개별적으로 추적 (분산 애플리케이션 모니터링 가능)
 - | 컨테이너 상태에 대한 Metrics 수집 → 외부 서비스와 연계

Docker Container Monitoring

Prometheus (Linux)

- /etc/docker/daemon.json 파일 수정

```
{  
  "metrics-addr": "0.0.0.0:9323",  
  "experimental": true  
}
```

| Docker 재실행

| Docker Prometheus 실행

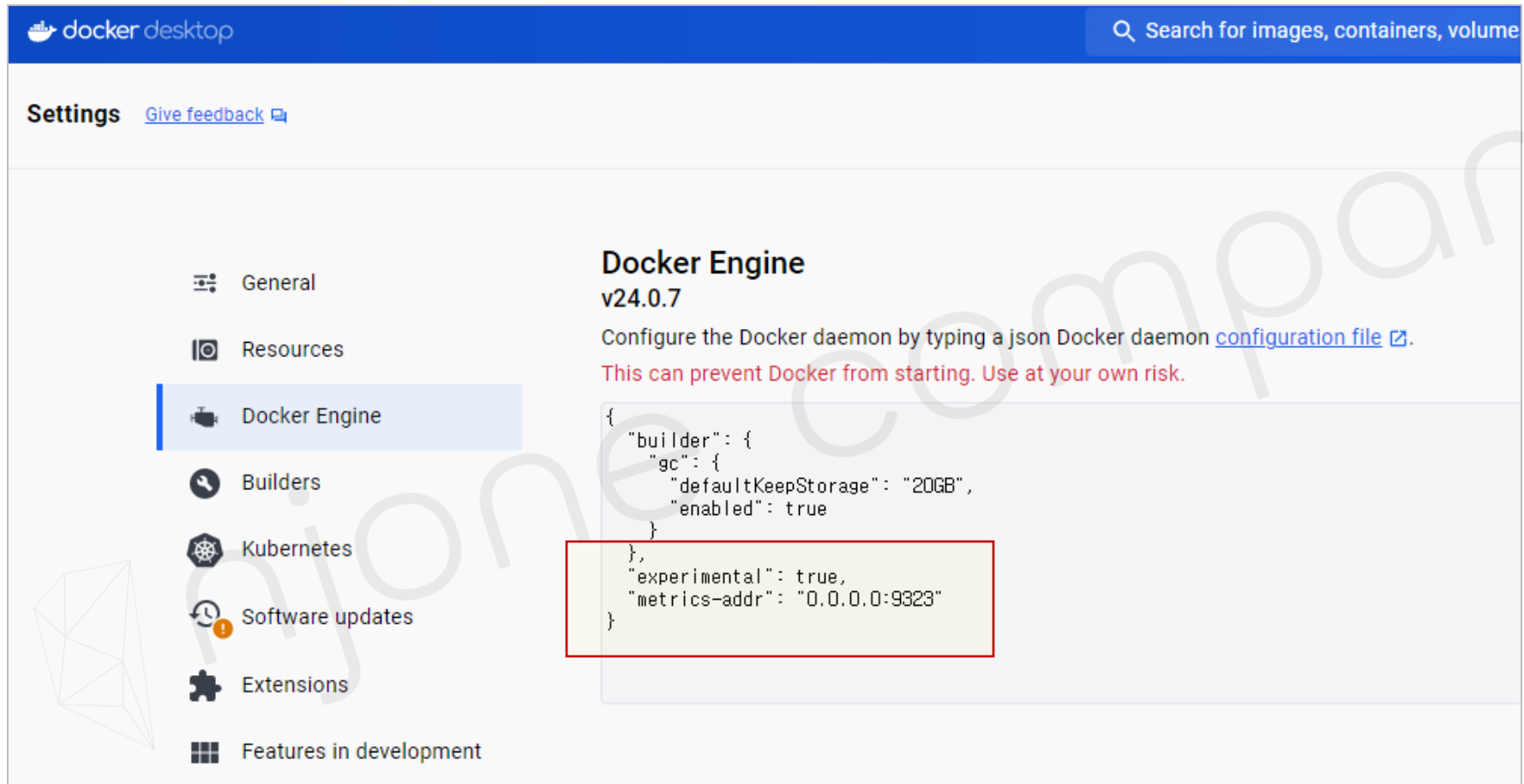
```
$ hostIP=172.31.5.111
```

```
$ docker container run -e DOCKER_HOST=$hostIP -d --rm -p 9090:9090 diamol/prometheus:2.13.1
```

Docker Container Monitoring

Prometheus (Windows & MacOS)

- Docker Desktop > Settings > Docker Engine → Docker Desktop 재실행



Docker Container Monitoring

Prometheus (Windows & MacOS)

- Docker Prometheus 실행

| Windows)

```
C:\> set hostIP=[IP Address]
```

```
C:\> docker container run -e DOCKER_HOST=%hostIP% -d --rm -p 9090:9090 diamol/prometheus:2.13.1
```

| MacOS)

```
$ hostIP=[IP Address]
```

```
$ docker container run -e DOCKER_HOST=$hostIP -d --rm -p 9090:9090 diamol/prometheus:2.13.1
```

Docker Container Monitoring

Prometheus

■ Prometheus 실행

```
← → ↻ ⚠ 주의 요약 43.203.199.147:9090/targets
← → ↻ ⚠ 주의 요약 43.203.199.147:9323/metrics

# HELP builder_builds_failed_total Number of failed image builds
# TYPE builder_builds_failed_total counter
builder_builds_failed_total{reason="build_canceled"} 0
builder_builds_failed_total{reason="build_target_not_reachable_error"} 0
builder_builds_failed_total{reason="command_not_supported_error"} 0
builder_builds_failed_total{reason="dockerfile_empty_error"} 0
builder_builds_failed_total{reason="dockerfile_syntax_error"} 0
builder_builds_failed_total{reason="error_processing_commands_error"} 0
builder_builds_failed_total{reason="missing_onbuild_arguments_error"} 0
builder_builds_failed_total{reason="unknown_instruction_error"} 0
# HELP builder_builds_triggered_total Number of triggered image builds
# TYPE builder_builds_triggered_total counter
builder_builds_triggered_total 0
# HELP engine_daemon_container_actions_seconds The number of seconds it takes to process each container action
# TYPE engine_daemon_container_actions_seconds histogram
engine_daemon_container_actions_seconds_bucket{action="changes",le="0.005"} 1
engine_daemon_container_actions_seconds_bucket{action="changes",le="0.01"} 1
engine_daemon_container_actions_seconds_bucket{action="changes",le="0.025"} 1
engine_daemon_container_actions_seconds_bucket{action="changes",le="0.05"} 1
```


Docker Container Monitoring

Prometheus

- Mariadb 컨테이너 실행

```
← → ↺ ⚠ 주의 요약 43.203.199.147:9323/metrics
engine_daemon_container_actions_seconds_bucket{action="start",le="0.25"} 1
engine_daemon_container_actions_seconds_bucket{action="start",le="0.5"} 6
engine_daemon_container_actions_seconds_bucket{action="start",le="1"} 6
engine_daemon_container_actions_seconds_bucket{action="start",le="2.5"} 6
engine_daemon_container_actions_seconds_bucket{action="start",le="5"} 6
engine_daemon_container_actions_seconds_bucket{action="start",le="10"} 6
engine_daemon_container_actions_seconds_bucket{action="start",le="+Inf"} 6
engine_daemon_container_actions_seconds_sum{action="start"} 1.5138992470000001
engine_daemon_container_actions_seconds_count{action="start"} 6
# HELP engine_daemon_container_states_containers The count of containers in various states
# TYPE engine_daemon_container_states_containers gauge
engine_daemon_container_states_containers{state="paused"} 0
engine_daemon_container_states_containers{state="running"} 2
engine_daemon_container_states_containers{state="stopped"} 0
# HELP engine_daemon_engine_cpus_cpus The number of cpus that the host system of the engine has
# TYPE engine_daemon_engine_cpus_cpus gauge
engine_daemon_engine_cpus_cpus 1
```

Docker Container Monitoring

Prometheus

- Graph

Prometheus Alerts Graph Status ▾ Help

☐ Enable query history

engine_daemon_container_states_containers{state="running"}

Execute

- insert metric at cursor · ⬆ ⬇ ⬇ ⬆

Graph

Console



Moment



Element

engine_daemon_container_states_containers{instance="DOCKER_HOST:9323",job="docker",state="running"}